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Reading in Children and Adolescents After Early Onset Hydrocephalus and in Normally Developing Age Peers: Phonological Analysis, Word Recognition, Word Comprehension, and Passage Comprehension Skill[1]

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Compared 50 children with early hydrocephalus, tested between the ages of 6 and 15 years, and their 51 age- and education-matched controls on 4 reading skills. Hydrocephalus and control groups did not differ in the ability to recognize words or to use phonological skills to decode "pretend" words, although the hydrocephalus group was poorer than the controls in understanding real words and texts. When individual differences in word recognition ability were con-

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trolled, the hydrocephalus group remained poorer than the controls in understanding single words and passages of text. Word recognition and passage comprehension skills were more highly correlated in the control group than in the hydrocephalus group. The passage comprehension scores of the hydrocephalus group lagged behind their word recognition scores, even for those children of normal or better IQ. Although the factors related to proficient and deficient reading comprehension for children with early hydrocephalus require further study, the present data show that adequate levels of word recognition, at least as measured by accuracy, in these children coexist with a significant deficit in reading comprehension.

KEY WORDS: reading; hydrocephalus; children; adolescents; normative development.

Reading and literacy are necessary to ensure an individual's full participation in society, and reading is important to scholastic achievement. A firm grasp of basic reading skills lays the foundation for the use of reading as a tool for learning subjects from literature and history to science and mathematics. The ability to decode individual words and the ability to understand the text are both necessary for reading mastery: Reading is impossible if the word code cannot be broken, or if the "alphabetic principle" is not learned, but the goal of reading is to derive meaning from text. Disturbances in the ability to decode single words automatically entails problems in school learning. Just as important, deficits in understanding what is read substantially disrupt school achievement.

A child who fails to acquire reading despite intact neurological status, normal intelligence, and unimpaired sensory-motor function, is usually deemed to be dyslexic. Dyslexic deficits are striking when they concern the acquisition of the written code, the ability to recognize single words. Some children with documented brain damage also have problems in learning to read single words, and these children will likely receive reading remediation. Other children with brain damage, however, may succeed in acquiring word recognition skills, but nonetheless remain poor readers. Such children have a more silent deficit: a problem in understanding what is read. Because their seemingly adequate word recognition serves to mask impaired comprehension, these children may fail to receive remedial help in reading.

Hydrocephalus is a disturbance of pressure within the brain that arises from an imbalance in the production and absorption of cerebrospinal fluid and that leads to engorgement of the cerebral ventricles and loss of brain volume, especially in the white matter (Harwood-Nash & Fitz, 1976; Hoffman, 1989). Early onset hydrocephalus, that diagnosed within the first year of life, is not an uncommon condition: Of the 1 in 500 live births with a neural tube defect, it is estimated that 85–90% will develop hydrocephalus. Hydrocephalus is also frequent in infants with postnatal complications such as intraventricular hemorrhage

(Hoffman, 1989). Although modern treatments have been successful in dramatically reducing the incidence of global mental retardation in children with hydrocephalus, such children remain vulnerable to specific cognitive impairment (see Fletcher & Levin, 1988). These invisible rather than visible handicaps exert a negative effect on the course of scholastic attainments and school achievement (e.g., Shaffer, Friedrich, Shurtleff, & Wolf, 1985; Tew & Laurence, 1972).

Hydrocephalus appears related to problems in reading mastery (Zeiner, Prigatano, Pollay, Biscoe, & Smith, 1985). When reading problems are demonstrated to exist, they appear to be associated with the presence of hydrocephalus that requires treatment, rather than to correlated conditions such as spina bifida (Tew & Laurence, 1975). But what kinds of reading problems are experienced by children with hydrocephalus? Children with hydrocephalus score below their same-age peers on global measures of reading (Zeiner et al., 1985, but see also Prigatano, Zeiner, Pollay, & Kaplan, 1983), and these global reading scores remain deficient even when adjusted for intelligence (Tew & Laurence, 1978). Global measures of reading ability, however, do not reveal which aspects of reading are deficient in this population and which, if any, are preserved. The literature dealing with more specific aspects of reading ability, such as single word recognition and comprehension, yield somewhat mixed results.

Some studies report the word recognition skills of children with hydrocephalus to be similar to those of their same-age peers (e.g., Anderson, 1973; Prigatano et al., 1983; Shaffer et al., 1985), whereas other studies find that these children have poorer word identifications skills than their peers (e.g., Halliwell, Carr, & Pearson, 1980; Tew & Laurence, 1975; Wills, Holmbeck, Dillon, & McLone, 1990). Some of the apparent discrepancy in these findings is likely due to differences in the selection of control groups (e.g., whether matched on IQ or not; whether controls are from the same educational system as the children with hydrocephalus, or whether comparisons are made to standardized norms), the sensitivity of the particular reading tasks used, the size of the groups, and the composition of the group with hydrocephalus with respect to formative pathology.

The information that exists on reading comprehension in hydrocephalus suggests that these children tend to have poorer comprehension skills than expected on the basis of their word recognition skills (Anderson, 1973; Halliwell et al., 1980). Further, studies of auditory-verbal comprehension show that some children with hydrocephalus have poor comprehension monitoring abilities (Dennis, Hendrick, Hoffman, & Humphreys, 1987), which suggests that they may be at risk as well for problems in comprehension and/or comprehension monitoring in the visual-verbal domain.

An analysis of cognitive functioning requires that global skills such as reading be studied with respect to their constituents, some of which may be relatively intact, others of which may be impaired. To date, there is little information on which aspects of the reading process are intact and which are deficient in children and adolescents with hydrocephalus. A componential analysis of reading is necessary if the process itself, or its pertubations, are to be understood.

What are some of the constituents of reading? Three constituents are considered in the componential analysis described here: The ability to recognize words or provide a natural reading of words without respect to meaning; the ability to use knowledge of phonology to decode new or less familiar words; and the ability to derive meaning from what is read.

The ability to recognize words quickly and accurately out of context discriminates good from poor comprehenders (Calfee & Piontkowski, 1981; Lesgold, Resnick, & Hammond, 1985). In beginning readers, letter and digit naming speed are related to later word recognition ability (Denckla & Rudel, 1974; Spring & Davis, 1988). Automatic word recognition may affect reading comprehension by freeing the reader to attend to understanding what is being read (LaBerge & Samuels, 1974; Perfetti, 1985). To be sure, word recognition does not always predict comprehension: College readers with a childhood diagnosis of dyslexia have relatively good reading comprehension ability despite inaccurate and slow word recognition (Bruck, 1990). In the normally developing reader, however, fast and accurate word recognition skills seem to coexist with proficient comprehension.

The role of phonological skill in reading acquisition and skilled reading is a matter of current exploration both from academic and pedagogical perspectives (Adams, 1990; Liberman & Liberman, 1990; Sawyer & Fox, 1991). Phonological skills differentiate good from poor readers (Backman, Bruck, Hebert, & Seidenberg, 1984; Share, Jorm, MacLean, & Matthews, 1984). Phonemic awareness, an early phonological skill conferring the knowledge that words are made up of units of sound, seems strongly related to early reading success. For example, sensitivity to rhyme (Bradley & Bryant, 1983; Bryant, MacLean, & Bradley, 1990; Kirtley, Bryant, MacLean, & Bradley, 1989) and the ability to blend phonemes to make words (Perfetti, Beck, Bell, & Hughes, 1987) each predict later reading success. Such phonemic awareness has been found to be a necessary, but not sufficient, condition for early reading success (Fox & Routh, 1984; Tunmer & Nesdale, 1985). These phonemic awareness skills may be necessary for the development of other phonological skills related more directly to beginning reading. For example, the ability to connect the phonemes with letters of the alphabet seems causally related to early word recognition and spelling (Ball & Blachman, 1991). Also, knowledge of spelling–sound correspondences may empower beginning readers to decode words that they have not seen before but that are part of their spoken vocabulary (Ehri & Wilce, 1985). Readers having more experience with the written word rely increasingly less on phonology as a means to accomplish word recognition (Backman et al., 1984; Seidenberg, Waters, Barnes, & Tanenhaus, 1984), even though young children use phonology to recognize many words (Backman et al., 1984), and phonology continues to be used by fluent adult readers to decode new words and to recognize less frequently seen words (Seidenberg et al., 1984).

Reading comprehension involves several skills, such as the ability to understand individual words, the ability to make pronomial reference, the ability to draw inferences, the ability to understand figurative language, the ability to derive the gist of a text, and the ability to monitor one's own understanding of what is being read so that comprehension failures can be detected and remedied. One important relationship that seems to hold over the whole course of development is that between vocabulary knowledge for individual words and the ability to understand passages of text (McKeown, Beck, Omanson, & Perfetti, 1983). The relationship may be reciprocal: Increasing vocabulary leads to improvements in text comprehension (McKeown et al., 1983), and word meanings are often learned from the reading context (Nagy & Anderson, 1984). Knowledge of how to monitor and correct comprehension failures is related to successful comprehension in older children (Paris & Oka, 1986).

The skills that account for reading success at one age may be different from those that account for reading success at another age. For example, phonological abilities account for more of the variance in reading ability in younger readers than in older readers (see Stanovich, 1986, for a review), whereas Verbal IQ accounts for little of the variance in reading at a young age, and for considerably more at an older age (Stanovich, Cunningham, & Feeman, 1984). Such findings suggest that some reading skills may be developmentally limited; that is, a skill (e.g., phonological recoding) may account for reading success at one point in development but not at another (Stanovich, 1986). The data also suggest that two skills may be reciprocally related such that improvements in one may lead to improvements in the other and vice versa (McKeown et al., 1983; Perfetti et al., 1987; Stanovich, 1986). Such findings also underscore the importance of studying reading skills over a wide age range.

What need study are the components of reading, and how these components change with development. With respect to the condition of hydrocephalus, little is known about the components of the reading process. Word recognition skills seem to be age-appropriate in some cases although not in others, and comprehension may tend to lag behind word recognition skill.

Little is also known about age-related skill changes in children and adolescents with hydrocephalus. Some studies of reading in hydrocephalus have used a single age group (e.g., Tew & Laurence, 1978), whereas others have used a wide age range but failed to analyze how age affects reading (e.g., Shaffer et al., 1985). Reading single words appears to improve with age in children with hydrocephalus just as in normally developing children (Wills et al., 1990). The comparisons in the Wills et al. study were made to standardized norms rather than to a control group of children with educational backgrounds similar to the children with hydrocephalus; to date, no other analyses of reading and age have been conducted. The decreasing ability of children with hydrocephalus to keep pace with their peers in some oral language skills (see Dennis et al., 1987), coupled with the fact that certain components of reading may be developmentally limited (Stanovich, 1986), suggest that the relationship between age and growth in reading skill for children with hydrocephalus and in their normally developing age peers needs to be studied.

The present study investigates four components of reading in children and adolescents with a history of early hydrocephalus. We explore the ability of 50 children with hydrocephalus ages 6 to 15, and that of 51 control children matched for age and education, to perform the following reading tasks: Give a natural reading for real English words, regardless of their meaning; use phonological knowledge to read "nonsense" or "pretend" words (most of which can be decoded using regular spelling-to-sound correspondence rules or analogies to similar sounding real words); understand the meaning of single written words; and understand passages of written text. The study explores a previously unanalyzed aspect of reading in children with hydrocephalus, phonological ability. It also investigates the understudied goal of reading, comprehension, more specifically than previously attempted. The study analyzes the ability to understand words read out of context, as well as the ability to understand passages of written text where text meaning draws on the meanings of individual words but also on other skills such as understanding figurative language, making pronomial reference, drawing inferences, and deriving the gist. The inclusion of five age groups (6–7, 8–9, 10–11, 12–13, and 14–15 years old) allows age-related changes in reading to be monitored.

METHOD

Subjects

The subjects were 50 children and adolescents with a history of early hydrocephalus and 51 age-matched controls without hydrocephalus. The controls were normally developing and normally achieving children recruited from the middle quartiles on the basis of their classroom performance, and in the same educational system as the children with hydrocephalus. Of the children with hydrocephalus, 41 were in the appropriate grade for their age and in the regular stream in that grade for most subjects; 6 were one grade behind but in the regular stream for that grade for most subjects; and 3 were in learning-disabled classrooms. Table I shows the five age groups used and the numbers of subjects with hydrocephalus and control subjects in each group. There were 26 boys and 25 girls in the hydrocephalus group, 19 boys and 32 girls in the control group.

Table I. IQ and Reading Test Performances in Children and Adolescents with Hydrocephalus and Controls[a]

Age (years)	n	VIQ	PIQ	Word identification	Word attack	Word comp.	Passage comp.
Hydrocephalic group							
6–7	9	96	95	58.0	38.2	55.3	43.8
		(72–131)	(73–120)	(28.9)	(34.5)	(35.4)	(28.9)
8–9	12	92	88	23.6	22.9	28.3	15.7
		(67–122)	(69–111)	(33.3)	(29.0)	(33.1)	(20.2)
10–11	10	89	82	32.4	29.4	29.2	16.0
		(69–108)	(49–101)	(31.3)	(26.4)	(25.3)	(15.2)
12–13	9	98	88	51.2	40.3	57.1	47.6
		(86–107)	(65–108)	(28.5)	(24.8)	(28.9)	(28 8)
14–15	10	94	91	55.3	47.8	48.1	36.9
		(70–120)	(70–120)	(30.1)	(30.1)	(24.6)	(27.0)
Control group							
6–7	9	—	—	59.8	35.6	66.2	45.8
				(34.9)	(27.6)	(32.6)	(28.1)
8–9	11	—	—	50.4	38.5	67.0	41.6
				(13.8)	(22.4)	(18.6)	(19.1)
10–11	11	—	—	45.1	50.9	60.9	51.3
				(22.1)	(17.2)	(15.8)	(25.8)
12–13	9	—	—	40.1	37.9	57.6	55.3
				(22.3)	(24.2)	(11 8)	(21.2)
14–15	11	—	—	61.1	57.9	60.5	65.5
				(31.5)	(21 5)	(24.1)	(23.4)

[a]Figures in the tables are grade percentiles (4 reading tasks) or standard scores (WISC-R). Ranges are presented with the IQ scores; standard deviations are presented with the reading scores.

The etiology of the hydrocephalus group is as follows: Myelodysplasia (n = 24), intraventricular hemorrhage (n = 9), congenital or ideopathic (n = 6), aqueduct stenosis (n = 5), Dandy-Walker cyst (n = 3), arachnoid cyst (n = 2), and choroid plexus papilloma (n = 1). These etiologies are those traditionally associated with classic early hydrocephalus, and they were not accompanied by other major central nervous system malformations such as agenesis of the corpus callosum. Admission to the study was based on the following criteria: diagnosis of hydrocephalus in the first year of life and the introduction of appropriate treatment after diagnosis, and either Verbal or Performance IQ above 70.

Measures

The Wechsler Intelligence Scale for Children–Revised (Wechsler, 1974) was administered to each subject with hydrocephalus. The reading assessment given to both hydrocephalus and control children comprised four subtests from the Woodcock Reading Mastery Test–Revised (WRMT-R; Woodcock, 1987): Word Identification, Word Attack, Word Comprehension, and Passage Comprehension. Each score from the WRMT-R took the form of a percentile rank based on grade.

Word identification requires that the child produce a natural reading for words taken from samples of reading material for different grades. This task measures the ability to identify words likely to be encountered below, at, and above the child's own grade level. It requires written word naming but no word comprehension. Accuracy, but not speed, is recorded.

Word attack requires that the child pronounce visually unfamiliar nonwords (e.g., "bipe") either by using tacit knowledge of the spelling–sound correspondence rules of English and/or by invoking previous experiences with similar-sounding real words. This task measures phonological ability as it is applied to written words, and indicates how the child has inferred the rules of English spelling–sound correspondence.

Word comprehension requires that the child read grade-appropriate words and produce antonyms (e.g., read "hot," produce "cold") and synonyms (e.g., read "cap," produce "hat") for them, as well as requiring the completion of analogies (e.g., read "dog–walks . . .bird ____" and produce "flies"). This task measures the understanding of semantic relations of single words that are not embedded in any text, and whose meaning does not rely on any text comprehension.

Passage comprehension requires that the child read silently and fill in the missing word from texts that range in complexity from easy ("The cat is playing with a ____" accompanied by a picture of a cat playing with a ball) to more difficult ("The fresh snow had been piled into a huge drift along the yard fence. Larry sank into it up to his waist but his father was able to work his way through the ____ and Larry followed in his path.") This task measures the ability to understand passages of text where comprehension may depend on vocabulary knowledge, grammatical knowledge, the ability to make pronomial reference, the ability to make an inference, and other cognitive-linguistic skills. To complete the missing word, the child must not only be able to read and understand the individual words in the passage but also be able to derive meaning from the text, using some of the abilities discussed above.

RESULTS

The intelligence test scores of the hydrocephalus sample were as follows: Verbal IQ (M = 93.5, SD = 14.9, range = 67–131); Performance IQ (M = 88.6, SD = 15.3, range = 49–120); and Full Scale IQ (M = 90.3, SD = 14.1, range = 62–129). Consistent with other studies (Dennis et al., 1981; Wills et al., 1990), Verbal IQ (VIQ) scores were significantly better than Performance IQ (PIQ) scores, t (48) = 2.24, p = .0295. The IQ scores for the different age groups are shown in Table I.

Reading and Intelligence

To understand whether reading skill is related to general cognitive function, the relationship between the four reading skills and intelligence was analyzed. Multiple regressions were conducted of Verbal IQ and Performance IQ on the four reading skills. IQ was significantly related to all four scores: The regression model for Word Identification, $F(2, 46) = 19.21$, $p = .0001$, accounted for 46% of the variance in Word Identification scores; the model for Word Attack, $F(2, 46) = 12.57$, $p = .0001$, accounted for 35% of the variance in Word Attack scores; the model for Word Comprehension, F (2, 46) = 19.77, $p = .0001$, accounted for 46% of the variance in Word Comprehension scores; and the model for Passage Comprehension, $F(2, 46) = 23.07$, $p = .0001$, accounted for 50% of the variance in Passage Comprehension scores. An analysis of the t statistics for the beta coefficients revealed a significant effect of VIQ but not PIQ for each reading measure: Word Identification, $t(46) = 5.69$, $p = .0001$; Word Attack, $t(46) = 4.93$, $p = .0001$; Word Comprehension, $t(46) = 4.98$, $p = .0001$; Passage Comprehension, $t(46) = 6.01$, $p = .0001$.

Although our sample did not include mentally retarded individuals (each subject had at least one IQ quotient of 70 or above), the range of intelligence was wider than would be found in a school classroom. Were a correlation of verbal intelligence and reading enhanced at the lower end of the IQ score distribution, then severely limited verbal intelligence would entail impairments in reading, but preservation of intelligence would not be predictive of reading ability. To evaluate this possibility, the regressions were performed for the subgroup with VIQ scores of 90 and above (n = 30). All four reading skills were still significantly related to IQ in this group, although to a lesser extent than when the full sample was included: The model accounted for 31% of the variance in Word Identification scores; for 23% of the variance in Word Attack scores; for 23% of the variance in Word Comprehension scores; and for 24% of the variance in Passage Comprehension scores. An analysis of the t statistics for the beta coefficients again revealed a significant effect of VIQ but not PIQ for each reading measure.

Reading and Early Hydrocephalus

Table I contains the mean percentile ranks of each reading task for each age group for the children with hydrocephalus and for the control children. Percentile ranks were assigned on the basis of school grade because this was deemed the best estimate of reading instruction. For the three children in the hydrocephalus group in ungraded special education classes, percentile ranks were based on one grade below the expected grade for their age.

A multivariate analysis of variance (MANOVA) was conducted on the four reading measures (Word Identification, Word Attack, Word Comprehension, and Passage Comprehension) comparing two levels of pathology (Hydrocephalus vs. Controls) and five age groups (6–7, 8–9, 10–11, 12–13, and 14–15 years old). There was a main effect of Pathology, $F = 6.025$, $p = .0002$, such that the hydrocephalus group performed more poorly than the controls, and a main effect of Age, $F = 2.715$, $p = .0005$. The interaction was not significant.

Univariate analyses were performed to explore these effects further. The first set of analyses concerns the effects of pathology. The hydrocephalus group was not significantly different from the control group on Word Identification and Word Attack. The two groups did differ, however, on Word Comprehension, $F(1, 91) = 12.935$, $p = .0005$, and on Passage Comprehension, $F(1, 91) = 17.314$, $p = .0001$. Now, grade-appropriate Word Comprehension and Passage Comprehension are contingent on having grade-appropriate Word Identification skills. Because measures of Word Comprehension and Passage Comprehension are not logically independent of the Word Identification measure, the question arises whether differences in Word Identification skill would account for differences in Word and Passage Comprehension for the two groups. To explore the source of the observed Word and Passage Comprehension deficits further, an analysis of covariance with Word Identification as the covariate was performed for Word Comprehension and Passage Comprehension separately with the two between-subjects factors of Level of Pathology and Age.

Accounting for Word Identification skill, the analysis for Word Comprehension revealed a significant effect for Pathology, $F(1, 81) = 3.998$, $p = .0489$, and a significant effect of Word Identification, $F(1, 81) = 57.362$, $p = .0001$. These results suggest that when differences in Word Identification skill are taken into consideration, the two groups still differ significantly in their ability to understand single words out of context. Covarying Word Identification skill, the analysis for Passage Comprehension revealed a significant effect of Pathology, $F(1, 81) = 5.251$, $p = .0245$, and a significant effect of Word Identification, $F(1, 81) = 47.961$, $p = .0001$. These results suggest that when differences in Word Identification skill are taken into consideration, the two groups still differ in their ability to understand passages of written text.

The next set of analyses concerns the effects of Age. In the univariate analyses, significant Age effects were present for Word Recognition, $F(4, 91) = 2.792$, $p = .0308$, and for Passage Comprehension, $F(4, 91) = 3.940$, $p = .0054$. There was a trend for Word Attack, $F(4, 91) = 2.088$, $p = .0888$, but no main effect of age for Word Comprehension. In the analyses of covariance, the main effect of Age was significant for Passage Comprehension only, $F(4, 81) = 2.739$, $p = .0342$.

Pairwise comparisons using the Duncan New Multiple Range Test ($p < .05$) revealed that the main effects of age for Word Identification concerned the poorer performance of children 8–9 and 10–11 years old compared to the performance of children 6–7 and 14–15 years old. The main effects of age for Passage Comprehension revealed relatively poorer performance of children 8–9 years old compared to those 6–7, 12–13, and 14–15 years old, and poorer performance of 10- to 11-year-old children compared to those 12–13 and 14–15 years old.

The Contribution of Reading Subskills to Passage Comprehension

Passage Comprehension is a global measure of text comprehension that requires the ability to recognize the words in the passage, the ability to decode words that are visually unfamiliar in order to ensure that comprehension is not disrupted, the ability to access the meanings of individual words, and likely several other skills including inferencing ability, the understanding of figurative language, and the ability to make pronomial reference. In this study, we address the contribution of the first three of these skills to one measure of global text comprehension: the Passage Comprehension task. And we compare the relative contribution that each of these skills makes to understanding text for the children with hydrocephalus and their controls. Age effects cannot be studied here because the groups are too small for a multivariate regression with three regressors. Simple regressions are also not appropriate to study age effects because the Word Identification and Word Attack measures must be correlated on a priori grounds with Word and Passage Comprehension; that is to say, the child must be able to recognize or decode the words on the Word Comprehension task before generating a synonym, an antonym, or an analogy, and must be able to decode the words on the Passage Comprehension task before providing a missing word in the text.

A multiple regression of Word Identification, Word Attack, and Word Comprehension on Passage Comprehension scores was performed, separately for the hydrocephalus and control groups. The model accounted for 72% of the variance in Passage Comprehension scores for the hydrocephalus group, $F(3, 46) = 38.516$, $p = .0001$, and for 45% of the variance in Passage Comprehension scores for the control group, $F(3, 47) = 12.721$, $p = .0001$. Do the two groups differ in terms of the size of the correlation between the reading subskills and Passage Comprehension? With sample sizes of 50 and 51 for the groups and a smaller R of .67 for the controls, the R for the hydrocephalus group must be greater than or equal to .83 in order to infer a significant difference at $p < .05$

between the two correlations (Millsap, Zalkind, & Xenos, 1990). Because the R for the hydrocephalus group is .85, therefore, the correlation coefficients from these two independent samples are deemed significantly different.

The t statistics for the beta coefficients from the multiple regressions revealed a significant effect of Word Identification, $t(46) = 2.174$, $p = .0349$, and Word Comprehension, $t(46) = 4.729$, $p = .0001$, for the hydrocephalus group. The pattern was somewhat different for the control group, with a significant effect of Word Comprehension, $t(47) = 2.471$, $p = .0172$, and a trend for Word Attack, $t(47) = 1.89$, $p = .0649$.

Do the Reading Comprehension Skills of Children with Hydrocephalus Lag Behind Their Word Recognition Skills?

One of the hypotheses generated from the literature on hydrocephalus and reading is that the comprehension skills of these children may lag behind relatively intact word recognition skills (Anderson, 1973; Halliwell et al., 1980). To test this hypothesis, a t test was conducted comparing Word Identification and Passage Comprehension scores for both children with hydrocephalus and controls. The Passage Comprehension scores for the hydrocephalus group were significantly poorer than their Word Identification scores, $t(49) = 3.989$, $p = .0002$, but the two skills did not differ for the control group. These data are shown in Table II.

A related question is whether the hydrocephalus children with VIQs of 90 and above also show significantly poorer Passage Comprehension than Word Identification, that is, is there something about a history of early hydrocephalus, even in the presence of intact intelligence, that disturbs particularly the comprehension of passages of text? The relevant means for this group are shown in Table II. The Passage Comprehension scores of these children were significantly poorer than their Word Identification scores, $t(29) = 2.896$, $p = .007$, following

Table II. Mean Reading Task Scores for the Full Sample of Children with Hydrocephalus, Children with Hydrocephalus with VIQ > 90, and for Controls

Task	Full hydrocephalus group (n = 50)		Hydrocephalus with VIQ > 90 (n = 30)		Controls (n = 51)	
	M	*SD*	*M*	*SD*	*M*	*SD*
Word identification	42.9	32.6	56.5	31.7	51.4	25.9
Word attack	35.1	29.3	46.3	29.1	44.8	23.4
Word comprehension	42.5	31.2	56.4	28.9	62.5	21.0
Passage comprehension	30.8	27.0	43.6	27.2	52.0	24.2

the pattern for the larger group. Thus, although this group of hydrocephalus children and adolescents with normal or better intelligence match age peers on several of the reading tasks other than passage comprehension, their weakness, like that of the entire hydrocephalus group, is in understanding passages of text.

DISCUSSION

Several core reading skills (providing a natural reading for words without respect to meaning; reading nonwords using phonological decoding skills; understanding words out of context; and understanding passages of text) were investigated in children and adolescents with early onset hydrocephalus and then compared to the development of these skills in age- and education-matched controls. The core reading skills were investigated in a number of ways by asking questions that arise both from the literature on reading in children with early hydrocephalus, as well as from the literature on reading acquisition in normally developing children: What is the level of reading in children with hydrocephalus when some of the core skills of reading are examined separately? Do the comprehension skills of children with hydrocephalus lag behind their word recognition skills? What is the relationship between IQ and reading in these children? Which subskills contribute to reading comprehension? Do the reading skills of children with hydrocephalus keep pace with those of their peers at each age point through development?

Two general issues emerge from an analysis of some of the components of reading in children with hydrocephalus compared to their normally developing peers. These concern the level of these reading skills and the reading skill pattern or configuration.

In terms of comparisons to the published grade norms, the children and adolescents with hydrocephalus performed at a low average to average level for their grade on the four reading skills measured in this study. In terms of comparisons with normally developing age peers in the same educational system, the children and adolescents with hydrocephalus performed similarly on measures of word recognition and phonological decoding skill. Their passage comprehension skills and their understanding of single words out of context were worse than those of their peers, even when differences in word recognition were accounted for.

The reading skills of children with hydrocephalus, then, are variable with respect to what aspect of reading is being measured, word decoding or comprehension. The distinctive and significant reading deficit found in children and adolescents with a history of early hydrocephalus is in comprehending what they read, not in decoding the printed words. This dissociation between accurate word decoding and poor comprehension skills in children with hydrocephalus may be contrasted with the opposite dissociation found in college adults with a childhood diagnosis of dyslexia. These are individuals who have intact comprehension skills in the face of relatively poor decoding skills (Bruck, 1990). Such dissociations between decoding and comprehension skills in special populations suggest that the strong positive relationship between decoding and comprehension skills that is usually found in normally developing readers may be of a fairly complex nature.

As measured here, phonological skills did not differ in children with hydrocephalus and controls. Although the phonological decoding skills of these children were at a fairly low level, these same skills in the normally developing controls were also lower than expected on the basis of the standardized norms. This may reflect the deemphasis on the explicit teaching of phonics in the educational system shared by both the children with hydrocephalus and the controls. Unlike language, reading is a skill that is intimately tied to instruction, especially in the early grades (Royer, Sinatra, & Schumer, 1990).

The finding that both children with hydrocephalus and controls had grade-based lower phonological analysis skills compared to the standardized norms, but that the two groups did not differ in this skill, underlines the necessity of using a control sample from the same educational or instructional system when investigating reading in pathological populations. Similarly, had the children with hydrocephalus been compared only to the standardized norms, their word comprehension scores might have appeared better than they actually were when compared to their peers who share the same history of reading instruction. The word comprehension skills of the controls were above that expected in comparison to the standardized norms. It is not clear why this should be the case; perhaps there is an emphasis in their shared educational system on vocabulary skills.

The reading skills studied here improved similarly with age for children with hydrocephalus and their normally developing peers. What remains to be seen is whether this is true of all aspects of reading comprehension. The test of textual understanding used in this study, Passage Comprehension, is an omnibus measure. Such global comprehension tasks fail to measure the separate contribution of skills such as inferencing and understanding of figurative language to overall reading comprehension.

The pattern of reading skill also differed in children and adolescents with hydrocephalus as compared to controls. The comprehension skills of the children with hydrocephalus were poorer than expected on the basis of their own levels of word recognition. The finding of impaired comprehension is consistent with the trends reported in the studies that have looked both at word recognition and comprehension skills (e.g., Anderson, 1973; Halliwell et al., 1980). We show additionally that this pattern held even in hydrocephalic children with IQs in the average to above average range. Despite good word recognition skills, these children had poorer passage comprehension skills. The findings suggest that early hydrocephalus leaves some aspects of reading unscathed, but affects the understanding of what is read, and that this negative effect on comprehension exists quite apart from any general cognitive or intellectual deficit.

Another difference in the pattern of reading skills between children with early hydrocephalus and their controls was in the nature of the skills that contributed to text comprehension and how well those skills accounted for understanding passages of text. When we knew how well children with hydrocephalus recognized words and understood words out of context, we could almost fully account for how well they would understand passages of text. In normally developing children, in contrast, the ability to understand individual words was important but much less directly predictive of text comprehension. Such findings suggest that in normally developing readers, unlike readers with congenital or perinatal brain damage, there are many skills in different combinations that may contribute to reading success.

Why is there a contribution of word recognition skills to text comprehension in children with early hydrocephalus but not in controls? Although the word recognition skills of these two groups were not different overall, more children with hydrocephalus had severe word recognition problems compared to the controls. What remains to be studied is how other skills not measured in this study contribute to the understanding of passages of text for children with hydrocephalus and their normally developing peers, and also, whether the constellation of skills that account for text comprehension change with development.

In dyslexic children without demonstrable brain injury, it is a matter of active debate whether and how IQ and reading are related (Leong, 1989; Lyons, 1989; Siegel, 1989). In normally developing children, IQ is a better predictor of reading skill at older than at younger ages (Stanovich et al., 1984). Verbal IQ and reading may be related at older ages in a reciprocal fashion; that is, a strong Verbal IQ aids understanding what is read, while reading, in turn, provides the opportunity to learn new vocabulary and other knowledge that contributes to the measure of Verbal IQ (Stanovich, 1986; Stanovich et al., 1984). In our study, Verbal IQ was strongly related to all four reading subskills, although it was less strongly related to reading in those children with IQs in the normal range. Even so, Verbal IQ cannot explain a great deal of the variance in reading in children with hydrocephalus, and even less so in children with hydrocephalus who are of average intelligence. It may be the case that the smaller relationship between IQ and reading holds only for the older children with hydrocephalus who have normal IQs as it does for older children without hydrocephalus, but there are insufficient children at each age in this normal IQ group to test this hypothesis.

As with other comparisons of IQ and language abilities in pathological groups with neurological damage, low IQ may entail poorer ability, but a normal IQ does not ensure that more specific cognitive abilities will achieve normal levels or follow normal patterns of development (Dennis & Barnes, 1990). Indeed, although the children with hydrocephalus who have a normal IQ perform at or above the average level on all the reading tasks, it is still the case that their text comprehension is poorer than that expected on the basis of their other reading skills.

The ability of children and adolescents to understand passages of text—the goal of most reading—needs to be studied further. Three areas seem especially important to the ability to understand what is read: the efficiency or speed and accuracy of word recognition, as opposed to accuracy alone; phonological skills other than phonological decoding ability; and general discourse skills.

Although the word recognition skills of children with hydrocephalus do not differ from those of controls, the measure used considers accuracy only. Efficiency of word recognition, which considers both speed and accuracy, may be a better predictor of reading skill. Some theories of reading fluency, in proposing that fast word level processes allow more cognitive resources to be dedicated to comprehension (LaBerge & Samuels, 1974; Perfetti, 1985), suggest one mechanism by which efficiency of word recognition is related to reading comprehension. No information on the efficiency of word recognition skills exists for children and adolescents with hydrocephalus. Slow but accurate word recognition skills, then, may be one important source of their problem in reading comprehension; the ability to integrate words into a message must suffer if some of the cognitive resources that are needed for integration are used, instead, for slow or labored word recognition. In young children, speed of access to names for color patches, letters, and digits is predictive of reading acquisition (Denckla & Rudel, 1974; Spring & Davis, 1988). Such speeded naming skills are worse in some children with hydrocephalus than in normally developing children (Dennis et al., 1987), so their fluency for recognizing real words may be compromised.

The research on the relationship between phonological skills and reading success in normally developing children bears on two questions concerning the development of reading in children with hydrocephalus: Does early phonological skill (phonemic awareness) predict early word recognition ability in children with hydrocephalus (i.e., does phonemic awareness discriminate between those children with hydrocephalus who seem to have no difficulty learning to read and those children with hydrocephalus who have problems acquiring word recognition skills) and how are these phonological skills in children with hydrocephalus related to their ability not only to learn to recognize words accurately but also to recognize words quickly?

Some views of reading acquisition posit that phonological skill is causally related to reading success (see Stanovich, 1986, for a review). Early phonological awareness skills (such as sensitivity to rhyme and the ability to parse the speech stream—a word—into its constituent sounds, and blend sounds into words) in preschoolers and beginning readers seem to be causally related to how quickly word recognition skills are acquired (Bradley & Bryant, 1983). What is unknown is whether special populations of children, such as those with hydrocephalus, develop early word recognition in a manner similar to that of normally developing children. Brain damage at an early age may have different consequences for the development of a skill, affecting the onset of the skill, the order in which skills or subskills emerge, and their relationship to one another (Dennis, 1988). The nature of the relationship between phonemic awareness and early reading success might be further illuminated by the extent to which the development of these early phonological skills and early word recognition ability in children with hydrocephalus parallels that in their normally developing peers. If these phonemic awareness skills are also precursors for reading success in children with early hydrocephalus, we might be able to target, at an early age, those children with early hydrocephalus who are at risk for later problems in the basics of reading–word recognition.

Phonemic awareness seems to allow the normally developing child to acquire spelling-to-sound knowledge, permitting the decoding of words, early in reading acquisition, on the basis of sound (Ehri & Wilce, 1985). Such spelling–sound knowledge has been shown to be important for acquiring word recognition skills (Backman et al., 1984). The present phonological task measured such spelling–sound knowledge, but because only accuracy was recorded, the task is unable to inform us about the speed with which children with hydrocephalus access spelling-to-sound knowledge. It might prove more fruitful, then, to ask how rapid access to spelling–sound knowledge is related to efficient word recognition, which, in turn, is related to good comprehension.

Such higher level phonological decoding skills, although seen as enabling the child to recognize words at the beginning stages of reading, rapidly come to be used less and less by young readers (Backman et al., 1984). This is not to say, however, that phonological skill does not contribute to fluent reading. It makes two important contributions to understanding what is read; it aids the recognition of new and less frequently seen words, and it fosters the integration of words into clauses so that meaning above the single word level can be derived. Phonology, though not used to accomplish the recognition of frequently read words, is used in the recognition of less frequently seen words (Seidenberg et al., 1984), and it is those less frequent words that tend to impart a disproportionate degree of meaning to the text (see Adams, 1990, for a discussion). Though not always used to accomplish word recognition, phonological information is likely always computed (Perfetti, Bell, & Delaney, 1988; Seidenberg & McClelland, 1989). This means that readily available phonological information may bolster the reader's memory for immediately preceding words and thus influence comprehension by aiding word integration (Baddeley, Vallar, & Wilson, 1987).

What is of interest, then, is whether in the early course of their reading children with hydrocephalus use spelling–sound information to accomplish word recognition as normally developing children do and also whether the phonological skills that seem importantly related to text comprehension in fluent readers are intact in children and adolescents with hydrocephalus. Such information can be obtained only from follow-up studies of the speed with which phonological information can be accessed.

The ability to understand written text does not just depend on efficient word recognition but also on several of the same skills that contribute to the comprehension of oral discourse: an adequate vocabulary; the ability to know when language is being used figuratively and what it means; the ability to deal with linguistic ambiguity; knowledge of grammatical structure; the ability to make pronomial reference; the ability to monitor one's own comprehension to detect and correct breakdown in understanding; and the ability to make inferences. Oakhill's work on a relatively small but interesting group of poor readers, those with no word recognition problems but substantial reading comprehension problems (Oakhill, 1982; Oakhill, Yuill, & Donaldson, 1990; Oakhill, Yuill, & Parkin, 1986) suggests that causal reasoning and inferencing are deficient in this group.

Hydrocephalus disrupts the development of several discourse skills. Dennis and Barnes (1992) found that children and adolescents with hydrocephalus have deficits in making inferences, manipulating linguistic ambiguity, producing speech acts, and understanding figurative language. These oral discourse skills are strongly related to how well these children understood passages of written text. In addition, the monitoring of oral language in some children and adolescents with hydrocephalus is poorer than that of their peers (Dennis et al., 1987). Taken together, such findings suggest that deficits in some or several discourse abilities might help to account for some of the written text comprehension problems experienced by children and adolescents with hydrocephalus.

The reading skills of individuals within any early hydrocephalus group are diverse. This variability means that some children have difficulty acquiring reading, while others acquire reading apace with their normally developing peers. There is a need to investigate how aspects of these children's medical histories might be predictive of a proficient or a deficient course of reading acquisition. Factors such as the etiology of the hydrocephalic condition, a history of shunt infection, and the presence of ocular defect have been found to be related to intelligence (Wills et al., 1990), language skill (Dennis et al., 1987), and word recognition skill (Tew & Laurence, 1978), suggesting that these and other medical history factors may also be predictive of performance on a number of reading skills. These issues are addressed in a companion paper.

In short, the present data provide evidence that some of the reading problems of children with hydrocephalus are silent, involving comprehension rather than the overt act of reading. However, many other important aspects of reading (e.g., phonological awareness and access to spelling–sound information, efficiency, or accuracy and speed of word recognition; general discourse skill and comprehension monitoring) need to be studied in these children to clarify the nature of their intact and impaired reading skills.

In our investigation of reading in children and adolescents with early onset hydrocephalus, we have explored several issues that we think are important for understanding a complex skill such as reading in any population with congenital or acquired neuropathology. A thorough understanding of reading proficiency and deficiency in such populations requires an analysis of the several component skills that contribute to reading success; identification of these skills derives from theories of reading acquisition in normally developing children. In addition, the investigation of reading in a pathological population must take the developmental parameters of the skill into account because studies of reading acquisition in normally developing children demonstrate that the constellation of skills that accounts for reading success at one age may be different from that that accounts for success at another age.

In outlining the direction we think the study of reading in hydrocephalus might profitably take, we have put forward a framework that need not be confined to this population. A theory-driven analysis of reading acquisition in several different pathological populations should contribute to a broader understanding of the skills that are important to both the normal acquisition and the pathological dissolution of reading.

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